

PERFORMANCE EVALUATION OF MULTI-ANTENNA MILLIMETER-WAVE COMMUNICATION SYSTEMS IN INDUSTRIAL SCENARIOS

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ABSTRACT: Future industrial wireless systems are expected to use millimeter wave (mmWave) frequency bands to achieve data rates beyond the capabilities of conventional sub-6 GHz systems. To guide the design and deployment of mmWave wireless systems, it is crucial to evaluate their performance under realistic conditions, using conventional sub-6 GHz as a reference. In this study, we assess achievable data rates through dynamic multi-band channel measurements, enabling a fair comparison between the two bands. Measurements are conducted in an indoor industrial environment at 2.55 GHz (sub-6 GHz) and 25.5 GHz (mmWave), with transmit antenna velocities of 50 km/h and 100 km/h. The results show that mmWave systems are highly sensitive to Doppler shifts, leading to significantly reduced performance in dynamic scenarios compared to sub-6 GHz channels.

Keywords: millimeter-wave, sub-6 GHz, measurements, industrial environments, performance evaluation

1. INTRODUCTION

Future industrial environments will rely on wireless sensor networks capable of supporting high data rates to enable remote control, automation and digitalization of industrial facilities [1]. In particular, advanced analytics and data-driven control enable automated and flexible industrial processes that can be monitored in real time, thereby reducing production downtime and maintenance costs. Compared to wired systems, wireless technology offers inherent advantages in terms of installation, flexibility, maintenance and network reconfiguration.

Current industrial wireless systems primarily operate in sub-6 GHz frequency bands, which provide relatively limited data rates due to spectrum shortage. A promising approach to overcome this limitation is the use of mmWave frequency bands (24 GHz – 300 GHz) [2], which offer substantially larger bandwidths than conventional sub-6 GHz systems [3, 4]. As a result, mmWave technology has gained significant attention not only in industrial environments [5, 6], but also in cellular networks [7, 8] and vehicular communications [9, 10, 11, 12, 13, 14]. Furthermore, mmWave bands are generally less congested than sub-6 GHz bands, which can simplify interference management [15].

Despite these advantages, mmWave communication introduces several challenges. Signal propagation at mmWave frequencies differs fundamentally from that at sub-6 GHz, exhibiting higher free-space path loss and increased sensitivity to blockage by obstacles. These characteristics can significantly limit communication range and reliability. Consequently, the design and deployment of mmWave-based industrial wireless systems require a careful performance evaluation relative to comparable sub-6 GHz systems under realistic operating conditions. Such an evaluation necessitates a precisely designed multi-band measurement setup and insights from real-world experiments are essential to guide the development and deployment of reliable mmWave communication systems for industrial applications.

Existing studies have investigated specific aspects of mmWave technology in industrial scenarios. For instance, [16] investigates the application of mmWave sensing for crack detection in packaged ceramic tiles, while [17] analyzes key physical-layer performance indicators of industrial communication using a 60 GHz mmWave antenna system. However, these works focus primarily on static scenarios and do not provide a systematic comparison with a sub-6 GHz baseline, which is essential for assessing the relative performance of mmWave systems.

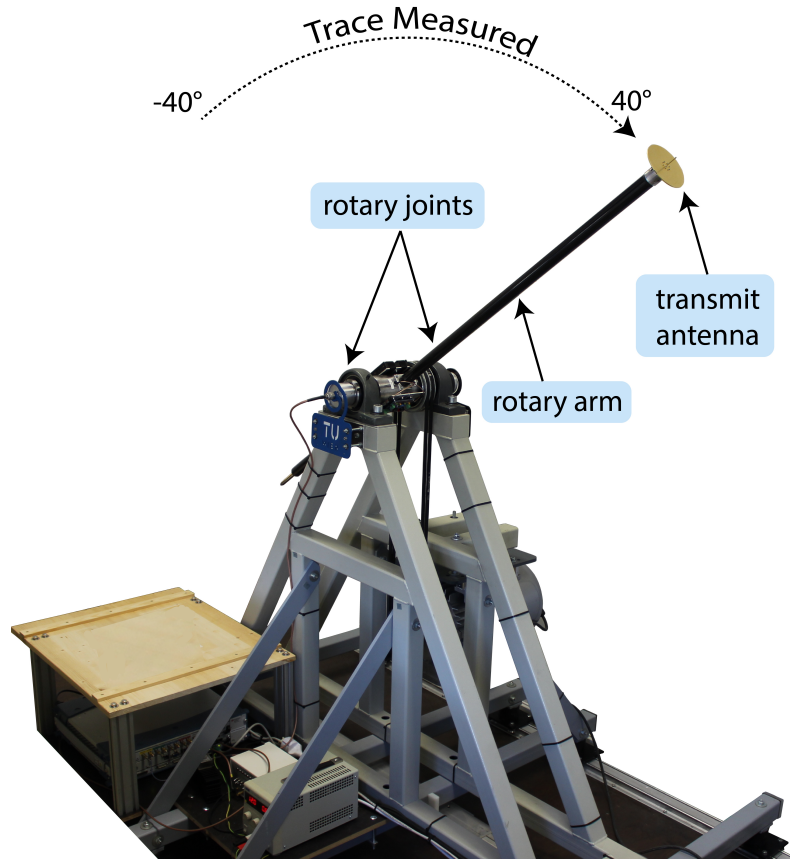


Fig. 1. The moving transmitter is based on the rotary unit, where either a sub-6 GHz or a mmWave antenna is mounted at the end of the rotating arm [18].

Contribution: In this paper, we evaluate the performance of a multi-antenna mmWave communication system using channel measurements conducted in a dynamic industrial environment. We enable a fair comparison with a conventional sub-6 GHz system using a multi-band measurement testbed introduced in [18]. We evaluate system performance in terms of achievable data rate as a function of the number of transmit antennas.

Organization: The remainder of this paper is organized as follows. In Section 2, we describe the measurement setup, the measurement campaign and the associated post-processing of the collected measurement data. Section 3 details the performance evaluation and discusses the results in terms of achievable data rate. Finally, Section 4 concludes the paper.

Notation: Scalars are denoted by x , vectors by bold lowercase letters $\mathbf{x}$ and matrices by bold uppercase letters $\mathbf{X}$. The superscripts $(\cdot)^T$ and $(\cdot)^H$ represent the transpose and Hermitian transpose, respectively, and $\|\cdot\|$ denotes the Euclidean norm.

2. DYNAMIC MULTI-BAND CHANNEL MEASUREMENTS

In this section, we describe the channel measurements conducted at sub-6 GHz and mmWave frequency bands in an indoor industrial environment. We first introduce the measurement setup, followed by a description of the measurement campaign and the post-processing applied to the collected data.

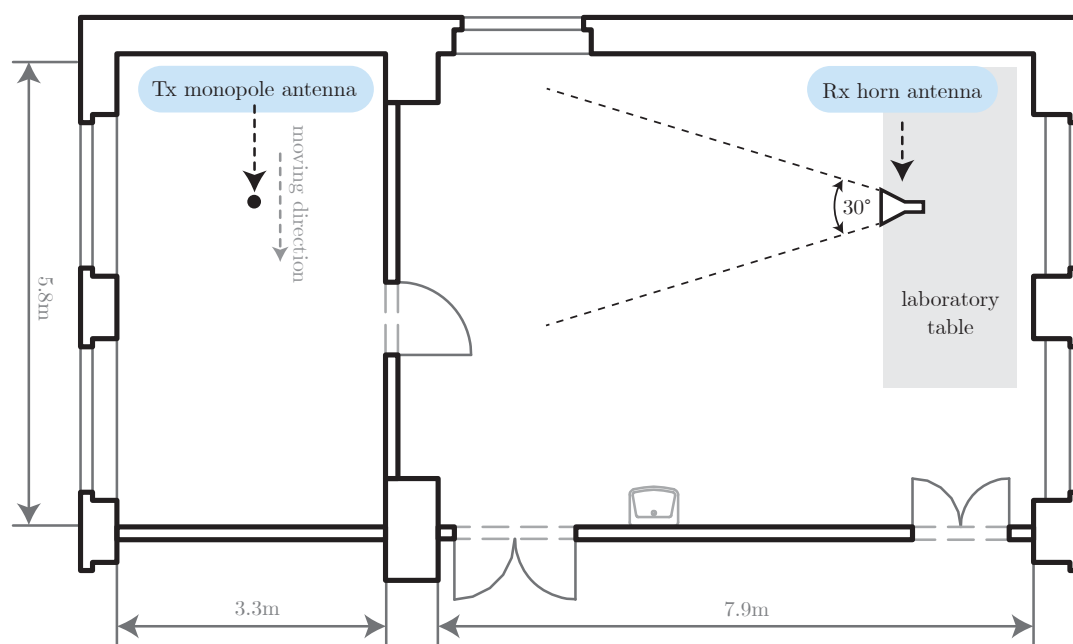


Fig. 2. Measured indoor environment [18] with the moving transmit antenna located in a smaller room and the static receive antenna placed in a larger room.

2.1. Measurement Setup

The considered measurement setup consists of a moving transmitter and a static receiver, and operates at carrier frequencies of 2.55 GHz (sub-6 GHz) and 25.5 GHz (mmWave). Transmitter mobility is realized using a rotary unit [19] that rotates an arm around a central axis at velocities of up to 400 km/h (see Fig. 1). The moving transmitter is equipped with a monopole antenna, which is attached to the end of the rotating arm, while the static receiver uses a horn antenna. Furthermore, the setup uses a trigger mechanism which initiates each measurement run at a precisely configurable arm angle, once per rotation. This design enables dynamic channel measurements with a well-defined and repeatable transmit antenna trajectory. In particular, identical motion traces of the transmit antenna and an identical receiver position are ensured for both frequency bands. As a result, the setup enables a fair and repeatable comparison of propagation and transmission characteristics between sub-6 GHz and mmWave systems. A detailed description of the measurement setup is provided in [18, 20, 21] and its accuracy has been validated in [22].

2.2. Measurement Campaign

Using the setup described in Section 2.1, channel measurements are conducted in an indoor environment, as illustrated in Fig. 2. The moving transmit antenna is mounted on the rotary unit and placed in a smaller room, while the static receive antenna is fixed on a laboratory table in a larger adjacent room. The separation between transmitter and receiver is approximately 7.5 m. This measurement scenario represents a private industrial wireless network operating at sub-6 GHz and mmWave frequencies. In this context, high mobility corresponds to fast-moving and rotating mechanical components of industrial machinery communicating with static control units. Measurements are performed at carrier frequencies of 2.55 GHz and 25.5 GHz, with transmit antenna velocities of 50 km/h and 100 km/h.

An orthogonal frequency-division multiplexing (OFDM) signal based on a Zadoff-Chu sequence [23] is transmitted while the transmit antenna rotates over an angular range from -40° to 40° at the rotary unit. At the receiver, we apply least-squares (LS) channel estimation to obtain the time-variant channel transfer function (CTF)

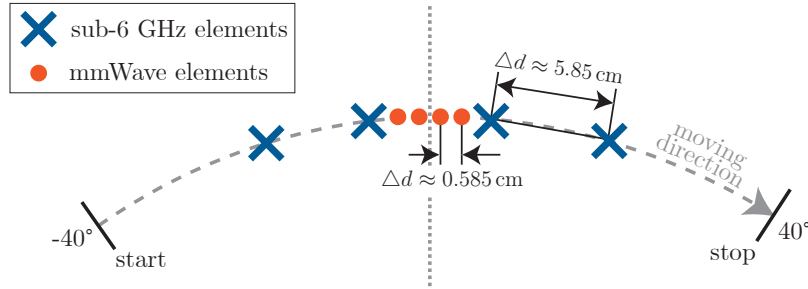


Fig. 3. Time-domain symbols interpreted as spatial samples forming a virtual transmit antenna array [18].

$H_{\text{meas}}[m, p]$, where $m \in \{-N_{\text{meas}}/2, \dots, N_{\text{meas}}/2 - 1\}$ denotes the subcarrier index and $p \in \{-K_{\text{meas}}/2, \dots, K_{\text{meas}}/2 - 1\}$ denotes the time-symbol index. Here, N_{meas} and K_{meas} represent the number of subcarriers and time-symbols used during the measurements, respectively. A detailed description of the measurement campaign and its parameters is provided in [18].

2.3. Post-Processing of Measurement Data

During the measurement campaign, subcarrier spacings of either $\Delta f_{\text{meas}} = 500 \text{ kHz}$ or $\Delta f_{\text{meas}} = 1 \text{ MHz}$ are employed, corresponding to symbol durations of $t_{s,\text{meas}} = 2 \mu\text{s}$ or $t_{s,\text{meas}} = 1 \mu\text{s}$, respectively. These subcarrier spacings are significantly larger than those typically used in practical sub-6 GHz and mmWave communication systems. To obtain time-variant CTFs with realistic subcarrier spacing, we apply sinc interpolation to the measured CTFs $H_{\text{meas}}[m, p]$ in both the time and frequency domains, yielding

$$H[n, k] = \sum_{p=1}^{K_{\text{meas}}} \sum_{m=1}^{N_{\text{meas}}} H_{\text{meas}}[m, p] \text{sinc}\left(\frac{t[k] - t_{\text{meas}}[p]}{t_s}\right) \text{sinc}\left(\frac{f[n] - f_{\text{meas}}[m]}{\Delta f}\right), \quad (1)$$

where $f[n] = (n - 1) \Delta f$ and $t[k] = (k - 1) t_s$, and similarly $f_{\text{meas}}[m] = (m - 1) \Delta f_{\text{meas}}$ and $t_{\text{meas}}[p] = (p - 1) t_{s,\text{meas}}$. The symbol duration of the interpolated CTF is denoted by t_s . This interpolation reduces the subcarrier spacing to $\Delta f = 60 \text{ kHz}$, corresponding to $t_s = 16.66 \mu\text{s}$. As a result, we obtain the interpolated time-variant CTF $H[n, k]$ for subcarrier indices $n \in \{-N/2, \dots, N/2 - 1\}$ and time-symbol indices $k \in \{-K/2, \dots, K/2 - 1\}$, where N and K denote the number of subcarriers and time-symbols, respectively.

3. PERFORMANCE EVALUATION

In this section, we evaluate system performance in terms of achievable data rate using the measured and interpolated time-varying CTFs at sub-6 GHz and mmWave frequencies. To emulate a multi-antenna transmitter, we model the moving transmit antenna as a virtual antenna array (VAA) by interpreting consecutive time-symbols as spatial samples [24], as illustrated in Fig. 3. Based on this approach, we construct a time-varying multiple-input single-output (MISO) channel by approximating an uniform linear VAA in the azimuth plane over multiple time-symbols, given by

$$\mathbf{h}[n, k'] = \mathbf{H}[n, k' + \Delta k \mathbf{d}] \in \mathbb{C}^{1 \times M_{\text{Tx}}}. \quad (2)$$

The antenna element positions are defined as

$$\mathbf{d} = \{0, 1, \dots, M_{\text{Tx}} - 1\} - M_{\text{Tx}}/2 + 0.5 \quad (3)$$

and the time-symbol index of the MISO channel is

$$k' \in \left\{ -\frac{K_{\text{eval}} - 1}{2}, \dots, \frac{K_{\text{eval}} - 1}{2} \right\}, \quad (4)$$

Table 1. Parameters Used for Performance Evaluation

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Subcarrier Spacing Δf	60 kHz	60 kHz
Transmit Power P_T	30 dBm	30 dBm
Wavelength λ	11.7 cm	1.17 cm
Evaluation Snapshots K_{eval}	31	31
Evaluation Bandwidth B_{eval}	20.04 MHz	400.02 MHz
Symbol Spacing Δk	240	24
Number of Transmit Antennas M_{Tx}	{2, 4, 8, 12}	{2, 4, 8, 12}

where K_{eval} denotes the number of time-symbols used for performance evaluation. The spatial distance between two consecutive time-symbols is $\Delta d_s = l_m/K \approx 0.233$ mm, given an arc length of $l_m \approx 1.4$ m and a total of $K = 6000$ time-symbols. This spacing is significantly smaller than the desired half-wavelength antenna spacing. To obtain approximately half-wavelength spacing in the VAA construction, every $\Delta k = 240$ time-symbols are selected for the sub-6 GHz band, and every $\Delta k = 24$ time-symbols for the mmWave band. This processing ensures comparable effective antenna spacing across the two frequency bands.

We consider a link establishment procedure consisting of a training phase, followed by a data transmission phase [25]. Since we assume a MISO configuration with a single receive antenna, transmission is limited to a single spatial stream. During the training phase, we create the precoder using a single time-symbol

$$k'_T = -\frac{K_{\text{eval}} - 1}{2}, \quad (5)$$

while the remaining $K_{\text{eval}} - 1$ time-symbols are used for data transmission. The channel used for training is given by

$$\bar{\mathbf{h}}[n] = \mathbf{h}[n, k'_T]. \quad (6)$$

To achieve the highest data rate, we compute the precoding vector $\bar{\mathbf{f}}[n] \in \mathbb{C}^{M_{\text{Tx}} \times 1}$ using a matched filter, represented as

$$\bar{\mathbf{f}}[n] = (\bar{\mathbf{h}}[n])^H. \quad (7)$$

To establish an upper performance bound and eliminate the impact of antenna movement, we also consider an ideal precoding case with perfect channel state information (CSI), where the precoder is updated for each time-symbol k' as

$$\mathbf{f}[n, k'] = (\mathbf{h}[n, k'])^H. \quad (8)$$

Transmit power constraints are ensured as

$$\|\bar{\mathbf{f}}[n]\|^2 = P_T \quad (9)$$

and

$$\|\mathbf{f}[n, k']\|^2 = P_T \quad (10)$$

for all subcarriers and time-symbols. The achievable data rate (expressed in bits/s), averaged over N subcarriers and $K_{\text{eval}} - 1$ time-symbols, is given by

$$R = \frac{B_{\text{eval}}}{N(K_{\text{eval}} - 1)} \sum_{\forall n} \sum_{\substack{\forall k' \\ k' \neq k'_T}} \log_2(1 + \beta[n, k']), \quad (11)$$

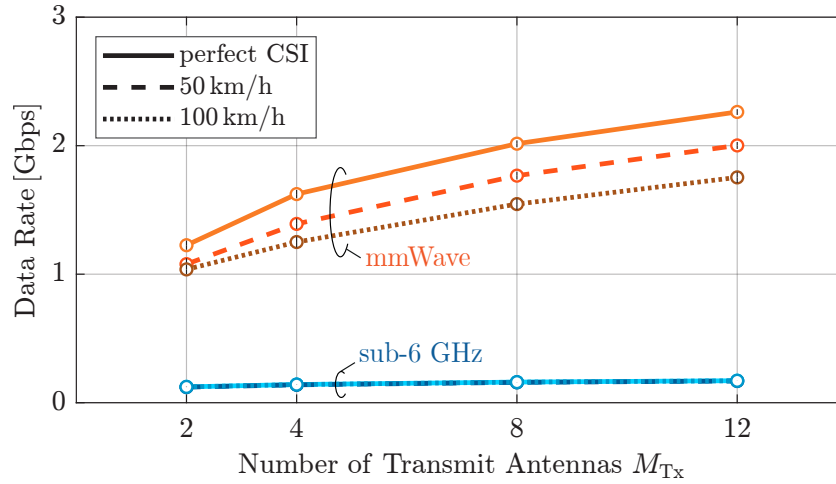


Fig. 4. The mmWave system achieves higher data rates than the sub-6 GHz system, but poses greater sensitivity to Doppler effects.

where B_{eval} denotes the evaluation bandwidth. The effective signal-to-interference-and-noise ratio (SINR) is given by

$$\beta[n, k'] = \frac{|\mathbf{h}[n, k'] \bar{\mathbf{f}}[n]|^2}{\bar{\sigma}_{wI}^2}, \quad (12)$$

with $\bar{\sigma}_{wI}^2$ representing the average noise-plus-interference power over the considered time-symbols,

$$\bar{\sigma}_{wI}^2 = \frac{1}{K_{\text{eval}}} \sum_{\forall k'} \sigma_{wI}^2[k']. \quad (13)$$

Details on the estimation of noise and interference are provided in [18].

In Fig. 4, we show the achievable data rate as a function of the number of transmit antennas, with evaluation parameters summarized in Table 1. We present results for imperfect precoding $\bar{\mathbf{f}}[n]$ at transmit antenna velocities of 50 km/h and 100 km/h for both frequency bands. Additionally, we show results for the perfect CSI case, which uses the ideal precoder $\mathbf{f}[n, k']$ for each time-symbol. To reflect realistic deployment scenarios, we use evaluation bandwidths compliant with 3rd Generation Partnership Project (3GPP) specifications [2], namely $B_{\text{eval}} = 20.04$ MHz for sub-6 GHz and $B_{\text{eval}} = 400.02$ MHz for mmWave.

The mmWave band achieves significantly higher data rates (up to approximately 2.2 Gbps) compared to the sub-6 GHz band, which reaches up to about 171 Mbps. This gain is primarily due to the substantially larger available bandwidth. For both frequency bands, the achievable data rate increases with the number of transmit antennas. In the sub-6 GHz band, increasing the number of antennas from 2 to 12 yields a gain of approximately 47.5 Mbps. In the mmWave band, the corresponding gain is about 1 Gbps in the perfect CSI case, which reduces to 0.92 Gbps at 50 km/h and further to 0.71 Gbps at 100 km/h.

Moreover, increasing transmit antenna velocity leads to a reduction in the maximum achievable data rate. For sub-6 GHz, this degradation is marginal, decreasing to approximately 170 Mbps (0.5% loss) at 50 km/h and 168 Mbps (1.75% loss) at 100 km/h, indicating limited sensitivity to Doppler-induced channel variations. In contrast, the mmWave band exhibits a pronounced performance degradation. Specifically, the achievable rate drops from approximately 2.2 Gbps under perfect CSI to 2 Gbps (9% loss) at 50 km/h and 1.75 Gbps (20% loss) at 100 km/h. This behavior highlights the increased susceptibility of mmWave systems to rapid channel variations caused by mobility, highlighting their greater vulnerability to Doppler effects compared to sub-6 GHz systems.

4. CONCLUSION

In this paper, we present a performance evaluation in terms of achievable data rate based on dynamic wireless channel measurements in an indoor environment at sub-6 GHz and mmWave frequency bands. The mmWave system achieves significantly higher data rates than the sub-6 GHz system, primarily due to its substantially larger available bandwidth. However, mmWave system performance is more strongly affected by Doppler shifts, leading to a faster degradation in achievable data rate with increasing transmitter velocity compared to sub-6 GHz. This increased sensitivity to mobility represents a key limitation for mmWave systems in dynamic scenarios. Future work could mitigate these challenges by incorporating channel prediction, adaptive beamforming or advanced tracking algorithms.

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